Project 3

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Setting Up

```
## Load in dataset

library("here")
library("tidyverse")
library("tidytuesdayR")

if (!dir.exists(here("data"))) {
    dir.create(here("data"))
}

b_lyrics <- readRDS(here("data", "b_lyrics.RDS"))
ts_lyrics <- readRDS(here("data", "ts_lyrics.RDS"))
sales <- readRDS(here("data", "sales.RDS"))

rds_files <- c("b_lyrics.RDS", "ts_lyrics.RDS", "sales.RDS")
## Check whether we have all 3 files
if (any(!file.exists(here("data", rds_files)))) {
    ## If we don't, then download the data
    b_lyrics <-
        readr::read_csv("https://raw.githubusercontent.com/rfor
        09-29/beyonce_lyrics.csv")
    ts_lyrics <-
        readr::read_csv("https://raw.githubusercontent.com/rfor
        09-29/taylor_swift_lyrics.csv")
    sales <-
        readr::read_csv("https://raw.githubusercontent.com/rfor
        09-29/sales.csv")

    ## Then save the data objects to RDS files
    saveRDS(b_lyrics, file = here("data", "b_lyrics.RDS"))
    saveRDS(ts_lyrics, file = here("data", "ts_lyrics.RDS"))
    saveRDS(sales, file = here("data", "sales.RDS"))
}
```

Note: This code was taken directly from instructions, and rest of code is wrapped, HTML cant find break in the URL so its a scroll for this block

only (sorry about that!)

Part 1: Explore Album Sales

Part 1A:

```
glimpse(sales)
```

```
Rows: 48
Columns: 8
$ artist     <chr> "Taylor Swift", "Taylor Swift", "Taylor
Swift", "Taylor Swi…
$ title      <chr> "Taylor Swift", "Fearless", "Fearless",
"Fearless", "Fearle…
$ country    <chr> "US", "WW", "US", "AUS", "UK", "WW", "US",
"UK", "WW", "US"…
$ sales      <dbl> 5720000, 12000000, 7180000, 500000, 609000,
5000000, 469400…
$ released   <chr> "October 24, 2006", "November 11, 2008",
"November 11, 2008…
$ re_release <chr> "March 18, 2008[34]", "October 27,
2009[40]", "October 27, …
$ label      <chr> "Big Machine", "Big Machine", "Big Machine",
"Big Machine",…
$ formats    <chr> "CD, CD+G (karaoke), DVD (karaoke), LP,
digital download", …
```

```
sales_tidy <- sales |>
    ## Removes "extra stuff" from datesm then convert use
        stringr to get date (includes regular expression)
    mutate(
        released = str_remove(released, " \\([A-Z]+\\)\\[[0-
        9]+\\]"),
        released = mdy(released)
    ) |>
    ## Use forcats to create a factor called country
    mutate(country = fct_collapse(country,
        "World" = c("World", "WW"),
        "FRA"   = c("FRA", "FR")
    )) |>
    ## Convert sales to album sales in millions of dollars.
```

```
    mutate(sales = sales / 1000000) |>
    ## Keep only album sales from the UK, the US or the World
    filter(country %in% c("UK", "US", "World"))

## Auto print your final wrangled tibble data frame.
sales_tidy
```

```
# A tibble: 36 × 8
   artist       title        country  sales released
re_release  label formats
   <chr>        <chr>        <fct>    <dbl> <date>     <chr>
<chr> <chr>
 1 Taylor Swift Taylor Swift US       5.72  2006-10-24 March
18, … Big … CD, CD…
 2 Taylor Swift Fearless     World   12     2008-11-11 October
27… Big … CD, CD…
 3 Taylor Swift Fearless     US       7.18  2008-11-11 October
27… Big … CD, CD…
 4 Taylor Swift Fearless     UK       0.609 2008-11-11 October
27… Big … CD, CD…
 5 Taylor Swift Speak Now    World    5     2010-10-25 <NA>
Big … CD, CD…
 6 Taylor Swift Speak Now    US       4.69  2010-10-25 <NA>
Big … CD, CD…
 7 Taylor Swift Speak Now    UK       0.169 2010-10-25 <NA>
Big … CD, CD…
 8 Taylor Swift Red          World    6     2012-10-22 <NA>
Big … CD, CD…
 9 Taylor Swift Red          US       4.46  2012-10-22 <NA>
Big … CD, CD…
10 Taylor Swift Red          UK       0.693 2012-10-22 <NA>
Big … CD, CD…
# ℹ 26 more rows
```

Part 1B:

```
sales_us <- sales_tidy |>
    ## Keep only US states
    filter(country == "US") |>
    ## Get release years and go to whole number (source below)
    mutate(
        years_since_release = floor(interval(released,
        today()) / years(1))
```

```
    )

sales_us
```

```
# A tibble: 13 × 9
   artist       title          country sales released
re_release label formats
   <chr>        <chr>          <fct>   <dbl> <date>     <chr>
<chr> <chr>
 1 Taylor Swift Taylor Swift   US       5.72 2006-10-24 March
18,… Big … CD, CD…
 2 Taylor Swift Fearless       US       7.18 2008-11-11 October
2… Big … CD, CD…
 3 Taylor Swift Speak Now      US       4.69 2010-10-25 <NA>
Big … CD, CD…
 4 Taylor Swift Red            US       4.46 2012-10-22 <NA>
Big … CD, CD…
 5 Taylor Swift 1989           US       6.22 2014-10-27 <NA>
Big … CD, CD…
 6 Taylor Swift Reputation     US       2.3  2017-11-10 <NA>
Big … CD, CD…
 7 Taylor Swift Lover          US       1.08 2019-08-23 <NA>
Repu… CD, LP…
 8 Beyoncé      Dangerously i… US       5.1  2003-06-23 <NA>
Colu… CD, LP…
 9 Beyoncé      B'Day          US       3.61 2006-09-01 <NA>
Sony… CD, CD…
10 Beyoncé      I Am... Sasha… US       3.38 2008-11-14 <NA>
Musi… CD, CD…
11 Beyoncé      4              US       1.5  2011-06-24 <NA>
Park… CD, di…
12 Beyoncé      Beyoncé        US       2.51 2013-12-13 <NA>
Park… CD, CD…
13 Beyoncé      Lemonade       US       1.55 2016-04-23 <NA>
Park… CD/DVD…
# ℹ 1 more variable: years_since_release <dbl>
```

```
## Group by Artist, summarizing by most recent, oldest, and
        median years since release by artist
sales_us |>
    group_by(artist) |>
    summarize(
        most_recent = min(years_since_release),
        oldest      = max(years_since_release),
```

```
        median      = median(years_since_release)
    )
```

```
# A tibble: 2 × 4
  artist       most_recent oldest median
  <chr>              <dbl>  <dbl>  <dbl>
1 Beyoncé               10     23     16
2 Taylor Swift           7     19     14
```

Sources I used to help me complete this:

https://lubridate.tidyverse.org/reference/interval.html

https://www.rdocumentation.org/packages/base/versions/3.6.2/topics/Round

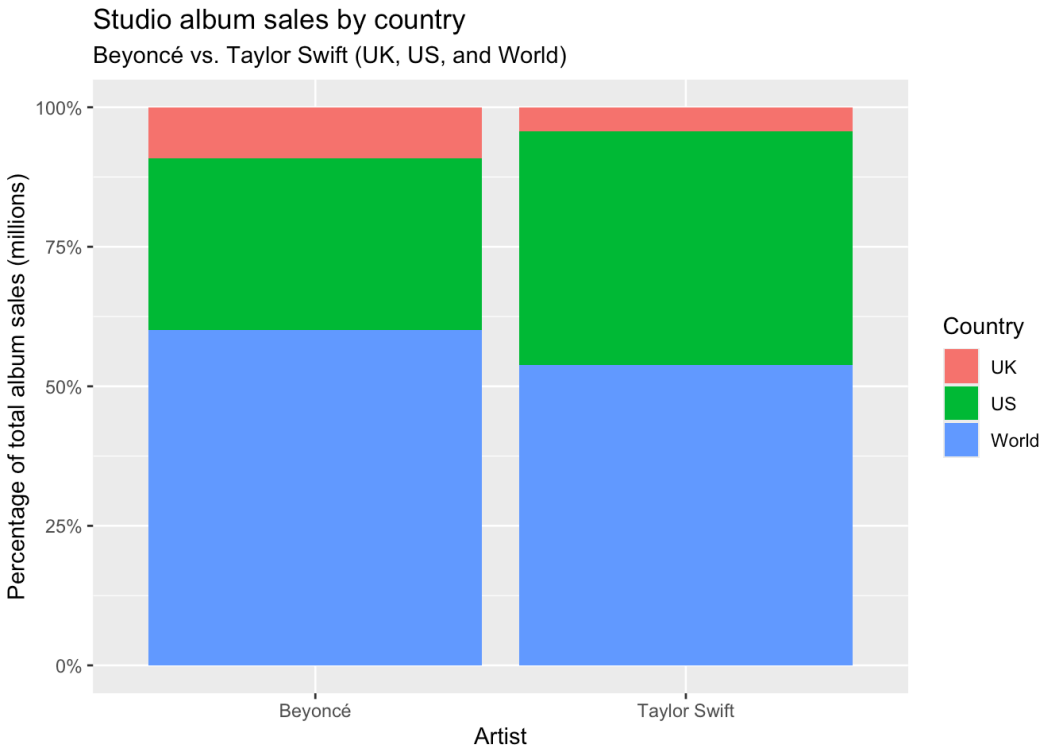
Part 1C

```
## Get total album sales in millions by artist and country
        using Part 1A data
total_sales <- sales_tidy |>
    group_by(artist, country) |>
    ## Added .groups = drops from error in console
    summarize(total_sales = sum(sales), .groups = "drop")
total_sales
```

```
# A tibble: 6 × 3
  artist       country total_sales
  <chr>        <fct>         <dbl>
1 Beyoncé      UK             5.24
2 Beyoncé      US            17.7
3 Beyoncé      World         34.5
4 Taylor Swift UK             3.32
5 Taylor Swift US            31.7
6 Taylor Swift World         40.8
```

```
##  Creating a percent stacked bar chart
total_sales |>
    ## Setting aes with x and y respectively, incorporating
        info for stacked bar chart
    ggplot(aes(x = artist, y = total_sales, fill = country)) +
    geom_bar(position = "fill", stat = "identity") +
    scale_y_continuous(labels = scales::percent) +
    ## Settinf labels
```

```
    labs(
        title = "Studio album sales by country",
        subtitle = "Beyoncé vs. Taylor Swift (UK, US, and
        World)",
        x = "Artist",
        y = "Percentage of total album sales (millions)",
        fill = "Country"
    )
```



Sources I used to help me complete this:

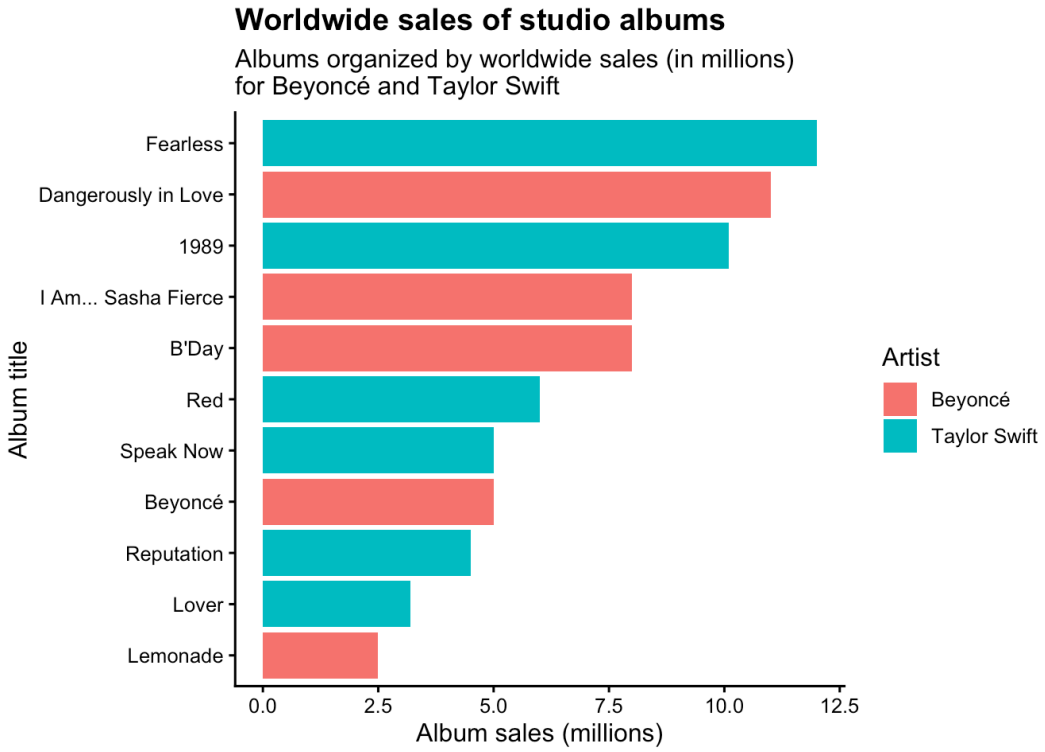
https://r-graph-gallery.com/48-grouped-barplot-with-ggplot2

https://ggplot2.tidyverse.org/reference/scale_continuous.html

Part 1D

```
## Data from 1A
sales_tidy |>
    ## Filtering for Wold
    filter(country == "World") |>
    ## including forcats functions
    mutate(title = fct_reorder(title, sales)) |>
```

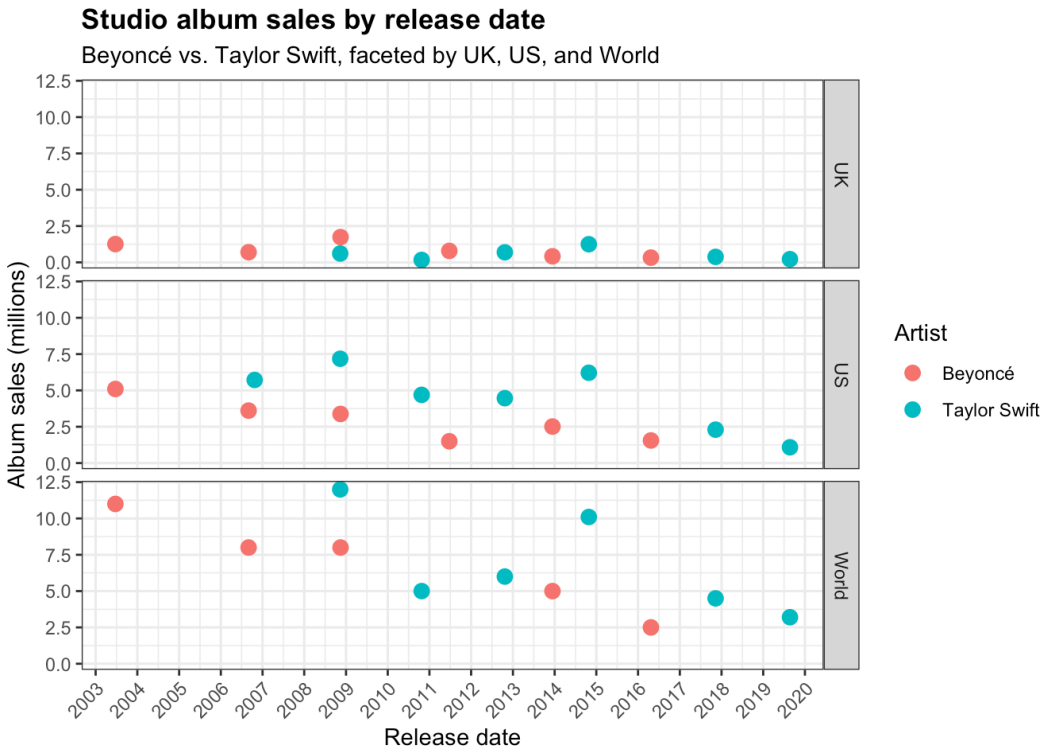
```
    ## Creating chart
    ggplot(aes(x = sales, y = title, fill = artist)) +
    geom_bar(stat = "identity") +
    ## Creating labels
    labs(
        title = "Worldwide sales of studio albums",
        subtitle = "Albums organized by worldwide sales (in
        millions)\nfor Beyoncé and Taylor Swift",
        x = "Album sales (millions)",
        y = "Album title",
        fill = "Artist"
    ) +
    theme_classic(base_size = 12) +
    theme(plot.title = element_text(face = "bold"))
```



Part 1E

```
## Data from Part 1A
sales_tidy |>
    ## Setting x, y, and color
    ggplot(aes(x = released, y = sales, color = artist)) +
    ## Setting point size
```

```
    geom_point(size = 3) +
    ## Using facet by rows (stacked)
    facet_grid(country ~ .) +
    scale_x_date(date_breaks = "1 year", date_labels = "%Y")+
    ## Setting labels
    labs(
        title = "Studio album sales by release date",
        subtitle = "Beyoncé vs. Taylor Swift, faceted by UK,
        US, and World",
        x = "Release date",
        y = "Album sales (millions)",
        color = "Artist"
    ) +
    theme_bw(base_size = 11) +
    theme(
        plot.title = element_text(face = "bold"),
        axis.text.x = element_text(angle = 45, hjust = 1)
        )
```



Sources I used to help me complete this:

https://ggplot2.tidyverse.org/reference/scale_date.html

Part 2: Exploring sentiment of lyrics

Part 2A

```
library("tidytext")

## One row per lyric line (keeps Album, Title, etc.)
ts_lines <- ts_lyrics %>%
    unnest_tokens(
        output = line,
        input = Lyrics,
        token = "lines",
        ## Keeping original capitalization for display
        purposes
        to_lower = FALSE
    )

## Filtering for "hello"
ts_hello <- ts_lines %>%
    filter(str_detect(line, regex("hello", ignore_case =
        TRUE)))

## Getting nrow of hellow
ts_hello %>% print(n = Inf)
```

```
# A tibble: 6 × 4
  Artist       Album    Title                 line
  <chr>        <chr>    <chr>                 <chr>
1 Taylor Swift Fearless Love Story            "And say,
\"Hello\""
2 Taylor Swift Red      I Almost Do           "That I can't
say \"Hello\" to y…
3 Taylor Swift Red      Everything Has Changed "'Cause all I
know is we said he…
4 Taylor Swift Red      Everything Has Changed "'Cause all I
know is we said he…
5 Taylor Swift Red      Everything Has Changed "All I know is
we said hello"
6 Taylor Swift Red      Everything Has Changed "All I know is
we said hello"
```

```
nrow(ts_hello)
```

```
[1] 6
```

```
## Filtering for goodbye "goodbye"
ts_goodbye <- ts_lines %>%
    filter(str_detect(line, regex("goodbye", ignore_case =
        TRUE)))

## Getting nrow of goodbye
ts_goodbye %>% print(n = Inf)
```

```
# A tibble: 12 × 4
   Artist       Album        Title                       line
   <chr>        <chr>        <chr>                       <chr>
 1 Taylor Swift Taylor Swift Tied Together With A Smile
"Goodbye, baby"
 2 Taylor Swift Speak Now    Mine                        "Braced
myself for the …
 3 Taylor Swift Speak Now    Back to December            "You
gave me all your l…
 4 Taylor Swift Speak Now    Long Live                   "And
force us into a go…
 5 Taylor Swift Red          I Almost Do                 "And
risk another goodb…
 6 Taylor Swift Red          Come Back Be Here
"Stumbled through the l…
 7 Taylor Swift 1989         All You Had to Do Was Stay "But
people like me are…
 8 Taylor Swift reputation   Getaway Car                 "Said
goodbye in "
 9 Taylor Swift reputation   Getaway Car                 "Said
goodbye in "
10 Taylor Swift Lover        Death By A Thousand Cuts    "Saying
goodbye is deat…
11 Taylor Swift Lover        Death By A Thousand Cuts    "'Cause
saying goodbye …
12 Taylor Swift Lover        Daylight                    "I'll
tell you truth, b…
```

```
nrow(ts_goodbye)
```

```
[1] 12
```

```
## Print line
```

```
cat("There are", nrow(ts_hello), "lines with \"hello\" and",
        nrow(ts_goodbye), "lines with \"goodbye\" in
        ts_lyrics.")
```

```
There are 6 lines with "hello" and 12 lines with "goodbye" in
ts_lyrics.
```

Part 2B

```
## Beyonce data already has one row per line, so we do not
        need to use unnest_tokens like we did above for
        ts_lyrics

## Filtering for "hello"
b_hello <- b_lyrics %>%
    filter(str_detect(line, regex("hello", ignore_case =
        TRUE)))

## Getting nrow for hello
b_hello %>% print(n = Inf)
```

```
# A tibble: 91 × 6
   line                         song_id song_name artist_id
artist_name song_line
   <chr>                          <dbl> <chr>         <dbl>
<chr>           <dbl>
 1 Hello world, well I just b… 2220711 "Dreamgi…       498
Beyoncé             5
 2 Hello Stevie, How you feel… 1981227 "Fingert…       498
Beyoncé             6
 3 Fellow great Americans, he… 2715227 "FREEDOM…       498
Beyoncé            52
 4 You had me at hello (Hello)   80249 "Hello"         498
Beyoncé            15
 5 Hello (Hello)                 80249 "Hello"         498
Beyoncé            16
 6 Hello (Hello)                 80249 "Hello"         498
Beyoncé            17
 7 You had me at hello (Hello)   80249 "Hello"         498
Beyoncé            18
 8 Hello (Hello)                 80249 "Hello"         498
Beyoncé            19
 9 Hello (Hello)                 80249 "Hello"         498
```

```
Beyoncé            20
10 'Cause you had me at hello…   80249 "Hello"         498
Beyoncé            24
11 Hello (Hello)                 80249 "Hello"         498
Beyoncé            25
12 Hello (Hello)                 80249 "Hello"         498
Beyoncé            26
13 You had me at hello (Hello)   80249 "Hello"         498
Beyoncé            41
14 Hello (Hello)                 80249 "Hello"         498
Beyoncé            42
15 Hello (Hello)                 80249 "Hello"         498
Beyoncé            43
16 You had me at hello (Hello)   80249 "Hello"         498
Beyoncé            44
17 Hello (Hello)                 80249 "Hello"         498
Beyoncé            45
18 Hello (Hello)                 80249 "Hello"         498
Beyoncé            46
19 You had me at hello (Hello)   80249 "Hello"         498
Beyoncé            50
20 Hello (Hello)                 80249 "Hello"         498
Beyoncé            51
21 Hello (Hello, hello)          80249 "Hello"         498
Beyoncé            52
22 You had me at hello (Hello)   80249 "Hello"         498
Beyoncé            62
23 Hello (Hello)                 80249 "Hello"         498
Beyoncé            63
24 Hello (Hello)                 80249 "Hello"         498
Beyoncé            64
25 You had me at hello (Hello)   80249 "Hello"         498
Beyoncé            65
26 Hello (Hello)                 80249 "Hello"         498
Beyoncé            66
27 Hello (Hello)                 80249 "Hello"         498
Beyoncé            67
28 You had me at hello (Hello)   80249 "Hello"         498
Beyoncé            71
29 Hello (Hello)                 80249 "Hello"         498
Beyoncé            72
30 Hello (Hello)                 80249 "Hello"         498
Beyoncé            73
31 You had me at hello           80249 "Hello"         498
Beyoncé            77
```

```
32 Hello                          80249 "Hello"         498
Beyoncé            78
33 Hello                          80249 "Hello"         498
Beyoncé            79
34 Hello                          80249 "Hello"         498
Beyoncé            80
35 You had me at hello (Hello) 4585886 "Hello (…       498
Beyoncé            14
36 Hello (Hello)               4585886 "Hello (…       498
Beyoncé            15
37 Hello (Hello)               4585886 "Hello (…       498
Beyoncé            16
38 You had me at hello (Hello) 4585886 "Hello (…       498
Beyoncé            17
39 Hello (Hello)               4585886 "Hello (…       498
Beyoncé            18
40 Hello (Hello)               4585886 "Hello (…       498
Beyoncé            19
41 'Cause you had me at hello… 4585886 "Hello (…       498
Beyoncé            23
42 Hello (Hello)               4585886 "Hello (…       498
Beyoncé            24
43 Hello (Hello)               4585886 "Hello (…       498
Beyoncé            25
44 You had me at hello (Hello) 4585886 "Hello (…       498
Beyoncé            39
45 Hello (Hello)               4585886 "Hello (…       498
Beyoncé            40
46 Hello (Hello)               4585886 "Hello (…       498
Beyoncé            41
47 You had me at hello (Hello) 4585886 "Hello (…       498
Beyoncé            42
48 Hello (Hello)               4585886 "Hello (…       498
Beyoncé            43
49 Hello (Hello)               4585886 "Hello (…       498
Beyoncé            44
50 You had me at hello (Hello) 4585886 "Hello (…       498
Beyoncé            48
51 Hello (Hello)               4585886 "Hello (…       498
Beyoncé            49
52 Hello (Hello, hello)        4585886 "Hello (…       498
Beyoncé            50
53 Hello                       2181490 "Intro –…       498
Beyoncé             1
54 You had me at hello (Hello) 2181490 "Intro –…       498
```

```
Beyoncé            18
55 Hello (Hello)              2181490 "Intro –…       498
Beyoncé            19
56 Hello (Hello)              2181490 "Intro –…       498
Beyoncé            20
57 You had me at hello (Hello) 2181490 "Intro –…       498
Beyoncé            21
58 Hello (Hello)              2181490 "Intro –…       498
Beyoncé            22
59 Hello (Hello)              2181490 "Intro –…       498
Beyoncé            23
60 Cuz you had me at hello (H… 2181490 "Intro –…       498
Beyoncé            26
61 Hello (Hello)              2181490 "Intro –…       498
Beyoncé            27
62 Hello oooooh               2181490 "Intro –…       498
Beyoncé            28
63 You had me at hello (Hello) 2181490 "Intro –…       498
Beyoncé            43
64 Hello (Hello)              2181490 "Intro –…       498
Beyoncé            44
65 Hello (Hello)              2181490 "Intro –…       498
Beyoncé            45
66 You had me at hello (Hello) 2181490 "Intro –…       498
Beyoncé            46
67 Hello (Hello)              2181490 "Intro –…       498
Beyoncé            47
68 Hello (Hello)              2181490 "Intro –…       498
Beyoncé            48
69 You had me at hello (Hello) 2181490 "Intro –…       498
Beyoncé            51
70 Hello (Hello)              2181490 "Intro –…       498
Beyoncé            52
71 Hello oooooh               2181490 "Intro –…       498
Beyoncé            53
72 You had me at hello (Hello) 2181490 "Intro –…       498
Beyoncé            63
73 Hello (Hello)              2181490 "Intro –…       498
Beyoncé            64
74 Hello (Hello)              2181490 "Intro –…       498
Beyoncé            65
75 You had me at hello (Hello) 2181490 "Intro –…       498
Beyoncé            66
76 Hello (Hello)              2181490 "Intro –…       498
Beyoncé            67
```

```
77 Hello (Hello)              2181490 "Intro –…       498
Beyoncé            68
78 You had me at hello (Hello) 2181490 "Intro –…       498
Beyoncé            71
79 Hello (Hello)              2181490 "Intro –…       498
Beyoncé            72
80 Hello oooooh               2181490 "Intro –…       498
Beyoncé            73
81 You had me at hello        2181490 "Intro –…       498
Beyoncé            76
82 Hello                      2181490 "Intro –…       498
Beyoncé            77
83 Hello ooooohuuu            2181490 "Intro –…       498
Beyoncé            78
84 Hello!                      299325 "Partiti…       498
Beyoncé            79
85 Hello!                      299325 "Partiti…       498
Beyoncé            91
86 Hello                      4481722 "Partiti…       498
Beyoncé            28
87 Hello                      4481722 "Partiti…       498
Beyoncé            31
88 Hello                      5838113 "Partiti…       498
Beyoncé            69
89 Bye-bye sadness, hello mel… 1848669 "So Amaz…       498
Beyoncé            12
90 You said hello to me        141848 "Yes"           498
Beyoncé            21
91 I said hello to you         141848 "Yes"           498
Beyoncé            22
```

```
nrow(b_hello)
```

```
[1] 91
```

```
# Filtering for "goodbye"
b_goodbye <- b_lyrics %>%
    filter(str_detect(line, regex("goodbye", ignore_case =
        TRUE)))

## Getting nrow for goodbye
b_goodbye %>% print(n = Inf)
```

```
# A tibble: 12 × 6
```

```
   line                         song_id song_name artist_id
artist_name song_line
   <chr>                          <dbl> <chr>         <dbl>
<chr>           <dbl>
 1 We only said goodbye with …  139043 Back to …       498
Beyoncé            12
 2 We only said goodbye with …  139043 Back to …       498
Beyoncé            21
 3 We only said goodbye with …  139043 Back to …       498
Beyoncé            24
 4 Thank God, I found the goo…   51492 Best Thi…       498
Beyoncé            38
 5 Thank God, I found the goo… 1946060 Best Thi…       498
Beyoncé            42
 6 Thank God, I found the goo… 4241137 Best Thi…       498
Beyoncé            38
 7 It's so hard to say goodbye  435491 Gift fro…       498
Beyoncé            23
 8 I never want to say goodbye  435491 Gift fro…       498
Beyoncé            24
 9 I never, ever want to say …  435491 Gift fro…       498
Beyoncé            25
10 We've got to say goodbye    1844620 Hard To …       498
Beyoncé            29
11 Don't have to say goodbye   1224115 Slow Love       498
Beyoncé            42
12 Somewhere between hi and g…  141848 Yes             498
Beyoncé            27
```

```r
nrow(b_goodbye)
```

```
[1] 12
```

```r
## Print line
cat("There are", nrow(b_hello), "lines containing \"hello\"
        and", nrow(b_goodbye), "lines containing
        \"goodbye\".")
```

```
There are 91 lines containing "hello" and 12 lines containing
"goodbye".
```

Part 2C

```
library(wordcloud)
?wordcloud

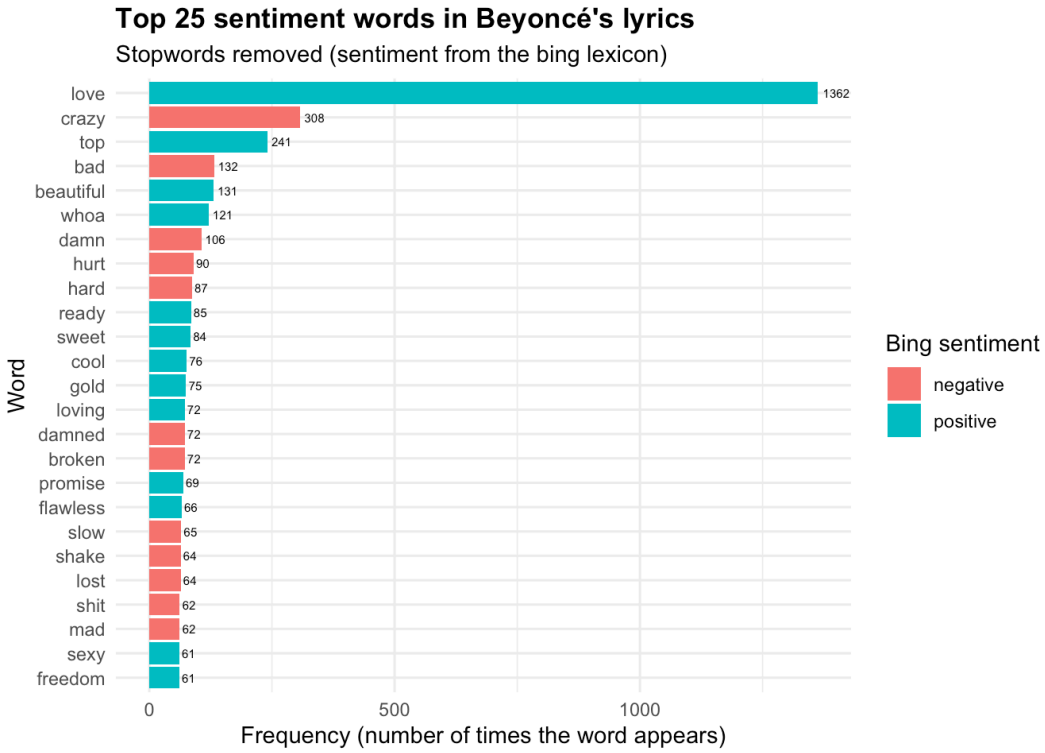
## Completing numbers 1-7
b_top25 <- b_lyrics %>%
    ## 1. One word per row
    unnest_tokens(output = word, input = line, token =
        "words") %>%
    ## 2. Removing stop words
    anti_join(stop_words, by = "word") %>%
    ## 3. Calculate the total number for each word in the
        lyrics
    count(word) %>%
    ## 4. Add column with Bing sentiment
    inner_join(get_sentiments("bing"), by = "word") %>%
    ## 5. Sort the rows from most frequent to least frequent
        words
    arrange(desc(n)) %>%
    ## 6. Top 25 most frequent words
    slice_head(n = 25)

## 7. Auto print the wrangled tibble data frame
b_top25
```

```
# A tibble: 25 × 3
   word          n sentiment
   <chr>     <int> <chr>
 1 love       1362 positive
 2 crazy       308 negative
 3 top         241 positive
 4 bad         132 negative
 5 beautiful   131 positive
 6 whoa        121 positive
 7 damn        106 negative
 8 hurt         90 negative
 9 hard         87 negative
10 ready        85 positive
# ℹ 15 more rows
```

```
## Creating the bar plot
b_top25 %>%
    mutate(word = fct_reorder(word, n)) %>%
    ## GGplot using aes setting x and y respectively
    ggplot(aes(x = n, y = word, fill = sentiment)) +
```

```
    geom_col() +
    ## Source for this below ; I decided to add so we can see
        exact values since x range is so vast
    geom_text(aes(label = n), hjust = -0.2, size = 2) +
    ## Setting labels
    labs(
        title = "Top 25 sentiment words in Beyoncé's lyrics",
        subtitle = "Stopwords removed (sentiment from the bing
        lexicon)",
        x = "Frequency (number of times the word appears)",
        y = "Word",
        fill = "Bing sentiment"
    ) +
    theme_minimal()+
    theme(plot.title = element_text(face = "bold"))
```




```
# Step 9: word cloud in seperate chunk for fig cap
```

```
# Step 9: word cloud
set.seed(1000)
b_top25 %>%
    with(wordcloud(word, n, max.words = 25, random.order =
        FALSE))
```



Word cloud of the top 25 sentiment words in Beyoncé's lyrics

Sources I used to help me complete this:

https://www.rdocumentation.org/packages/wordcloud/versions/2.6/topics/wordcloud

https://ggplot2.tidyverse.org/reference/geom_text.html

https://quarto.org/docs/authoring/figures.html

Part 2D

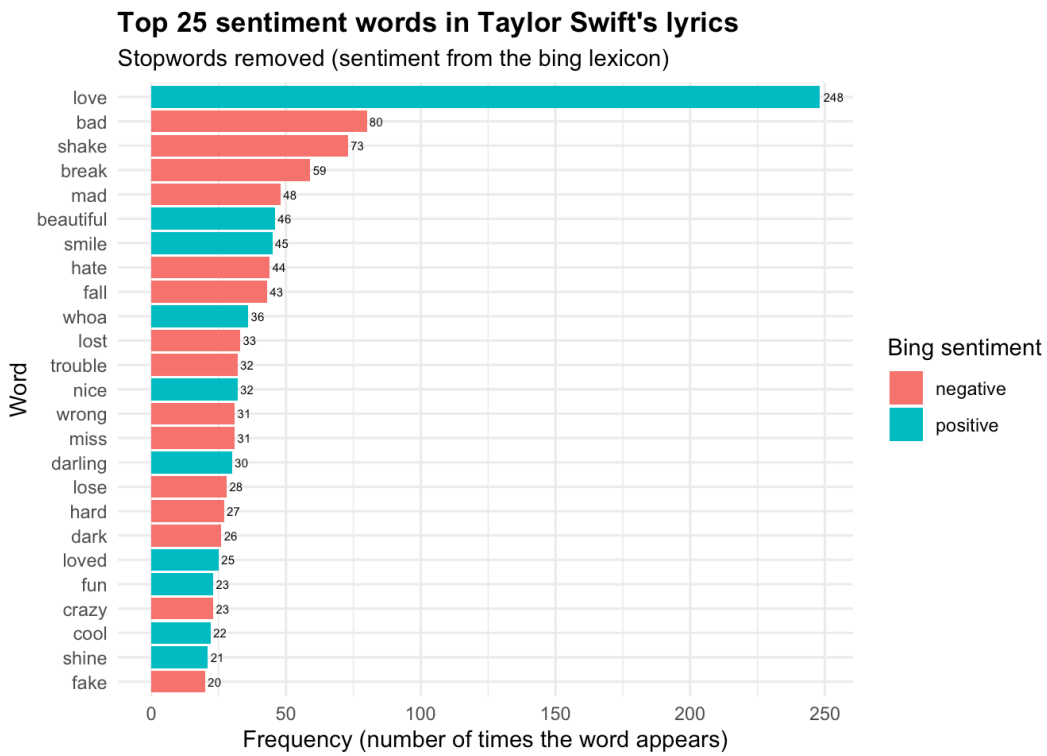
```
## Completing numbers 1-7
ts_top25 <- ts_lines %>%
    ## 1. One word per row (ts_lines from Part 2A has one
        lyric line per row)
    unnest_tokens(output = word, input = line, token =
        "words") %>%
    ## 2. Removing stop words
    anti_join(stop_words, by = "word") %>%
    ## 3. Calculate the total number for each word in the
        lyrics
    count(word) %>%
    ## 4. Add column with Bing sentiment
    inner_join(get_sentiments("bing"), by = "word") %>%
```

```
    ## 5. Sort the rows from most frequent to least frequent
        words
    arrange(desc(n)) %>%
    ## 6. Top 25 most frequent words
    slice_head(n = 25)

## 7. Auto print the wrangled tibble data frame
ts_top25
```

```
# A tibble: 25 × 3
   word          n sentiment
   <chr>     <int> <chr>
 1 love        248 positive
 2 bad          80 negative
 3 shake        73 negative
 4 break        59 negative
 5 mad          48 negative
 6 beautiful    46 positive
 7 smile        45 positive
 8 hate         44 negative
 9 fall         43 negative
10 whoa         36 positive
# ℹ 15 more rows
```

```
## Creating the bar plot
ts_top25 %>%
    mutate(word = fct_reorder(word, n)) %>%
    ## ggplot using aes setting x and y respectively
    ggplot(aes(x = n, y = word, fill = sentiment)) +
    geom_col() +
    ## Adding counts at the end of each bar (geom_text,
        ggplot2 documentation)
    geom_text(aes(label = n), hjust = -0.2, size = 2) +
    ## Setting labels
    labs(
        title = "Top 25 sentiment words in Taylor Swift's
        lyrics",
        subtitle = "Stopwords removed (sentiment from the bing
        lexicon)",
        x = "Frequency (number of times the word appears)",
        y = "Word",
        fill = "Bing sentiment"
    ) +
    theme_minimal()+
    theme(plot.title = element_text(face = "bold"))
```




```
# Step 9: word cloud in seperate chunk for fig cap
```

```
# Step 9: word cloud
set.seed(1000)
ts_top25 %>%
    with(wordcloud(word, n, max.words = 25, random.order =
        FALSE))
```



Word cloud of the top 25 sentiment words in TS's lyrics

Sources I used to help me complete this are same for Part 2C.

Part 2E

```
## Ran tidytext::get_sentiments("afinn") in console

## Completing Parts 1-6
ts_album_sentiment <- ts_lyrics %>%
    ## Clean album names (remove white space at end) so they
        match the sales data (source below)
    mutate(Album = str_to_title(str_trim(Album))) %>%
    ## 1. One word per row
    unnest_tokens(output = word, input = Lyrics, token =
        "words") %>%
    ## 2. Removing stop words
    anti_join(stop_words, by = "word") %>%
    ## 3. Calculate the total number for each word in the
        lyrics for each Album
    count(Album, word) %>%
    ## 4. Add column with afinn sentiment score
    inner_join(get_sentiments("afinn"), by = "word") %>%
    ## 5. Calculate the average sentiment score for each
        album, weighted by how often each word appears
```

```
    group_by(Album) %>%
    summarize(avg_sentiment = sum(value * n) / sum(n))

## 6. Auto print
ts_album_sentiment
```

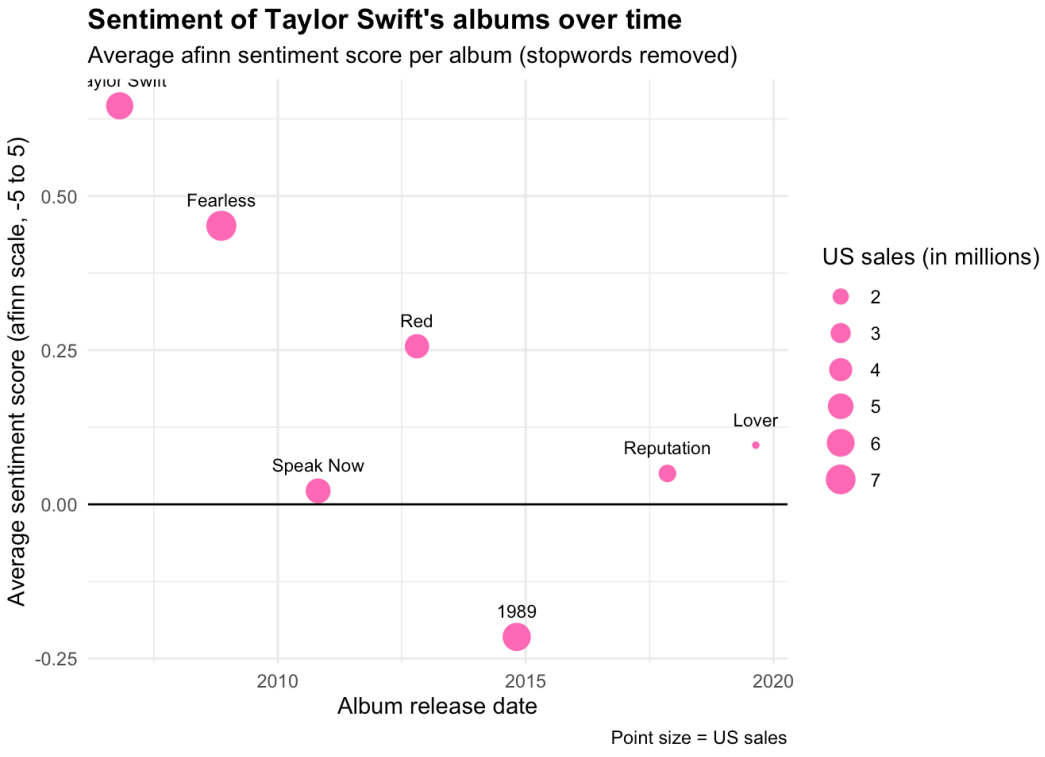
```
# A tibble: 8 × 2
  Album        avg_sentiment
  <chr>                <dbl>
1 1989               -0.215
2 Fearless            0.452
3 Folklore           -0.396
4 Lover               0.0960
5 Red                 0.256
6 Reputation          0.0501
7 Speak Now           0.0219
8 Taylor Swift        0.646
```

```
## 7. Join with Part 1A US sales
ts_sentiment_sales <- sales_tidy %>%
    filter(artist == "Taylor Swift", country == "US") %>%
    inner_join(ts_album_sentiment, by = c("title" = "Album"))

ts_sentiment_sales
```

```
# A tibble: 7 × 9
  artist   title country sales released   re_release label
formats avg_sentiment
  <chr>    <chr> <fct>   <dbl> <date>     <chr>      <chr>
<chr>           <dbl>
1 Taylor … Tayl… US       5.72 2006-10-24 March 18,… Big … CD,
CD…         0.646
2 Taylor … Fear… US       7.18 2008-11-11 October 2… Big … CD,
CD…         0.452
3 Taylor … Spea… US       4.69 2010-10-25 <NA>       Big … CD,
CD…         0.0219
4 Taylor … Red   US       4.46 2012-10-22 <NA>       Big … CD,
CD…         0.256
5 Taylor … 1989  US       6.22 2014-10-27 <NA>       Big … CD,
CD…        -0.215
6 Taylor … Repu… US       2.3  2017-11-10 <NA>       Big … CD,
CD…         0.0501
7 Taylor … Lover US       1.08 2019-08-23 <NA>       Repu… CD,
LP…         0.0960
```

```
## 8-10. Scatter plot
ts_sentiment_sales %>%
    ggplot(aes(x = released, y = avg_sentiment)) +
    geom_point(aes(size = sales), color = "hotpink") +
    ## 9. Horizontal line at 0 (neutral sentiment) SOurce
        below
    geom_hline(yintercept = 0, linetype = "solid") +
    ## Album names above each point (source same as above
        parts)
    geom_text(aes(label = title), vjust = -1.5, size = 3) +
    ## Adding labels to the graph
    labs(
        title = "Sentiment of Taylor Swift's albums over
        time",
        subtitle = "Average afinn sentiment score per album
        (stopwords removed)",
        caption = "Point size = US sales",
        x = "Album release date",
        y = "Average sentiment score (afinn scale, -5 to 5)",
        size = "US sales (in millions)"
    ) +
    theme_minimal() +
    theme(plot.title = element_text(face = "bold"))
```



Sources I used to help me complete this:

https://www.rdocumentation.org/packages/ggplot2/versions/0.9.1/topics/geom_hline

https://www.rdocumentation.org/packages/stringr/versions/1.6.0/topics/str_trim

10. Taylor Swift's albums have decreased in sentiment over time. Her first two albums (Taylor Swift and Fearless) have the highest average sentiment scores. The albums following that have had a relatively neutral sentiment score, except for 1989 which is negative. One could argue that Red has a slightly higher score, but it is still only around 0.25 on the graph. We can also see how sentiment does not necessarily track sales. 1989, having the lowest sentiment score was one of her best-selling albums, as was Fearless which had one of the highest scores.